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Energy Communities: Pervasive Technologies and Collective Futures

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Abstract—Energy communities are emergent sociotechnical constellations where local actors collectively organize green energy initiatives. Embedded in the vision of energy communities are expectations that pervasive energy technologies will foster citizen participation toward sustainable, democratic futures. Those technologies are predominantly designed as generalizable "one-size-fits-all" solutions to automate the coordination, production, consumption, and distribution of local green energy. However, these solutions rarely consider the diverse cultural and social practices of local communities in their designs. In this article, we draw on emerging research and five situated design cases to illustrate the many nuances involved in envisioning and developing technology for energy communities. Reflections from these cases suggest a need for design approaches that better account for the diverse social lifeworlds into which these technologies are meant to foster local engagement towards sustainable futures.

The concept of energy communities is increasingly gaining political and societal traction, particularly in a European context. The vision of energy communities is part of the European Union's strategy of bringing political support for communities to assemble in local green energy collectives to transition towards a more green, digitalized, and decentralized energy system [1]. The purpose of introducing the concept of energy communities at an EU level is to advance the development of *"energy technologies and to strengthen consumer participation in the energy markets whilst prioritizing the local production of electricity from renewable energy sources"* [2]. Embedded in this vision is a householder who willingly participates in the local energy community. Supported by digital tools and real-time price signals [1], this householder is expected to welcome pervasive energy technologies into their home as long they only foster *"minor inconveniences and discomforts in everyday life"* [3].

To help realize such futures, we see large-scale EU research and development projects exploring the potentials of digital data, computational algorithms, and automation for engaging EU citizens in communal green energy initiatives [2]. This work presents visions of future sociotechnical energy infrastructures typically promoted as generalizable "one-size-fits-all" solutions that aim to anchor green energy infrastructures within diverse social and cultural communities around Europe and beyond. A large corpus of projects has investigated the technical feasibility of decentralized community energy infrastructures, but little is known of how these may enhance democratic engagement, ensure energy justice, and foster long-lasting participation in local communities. Since the organization of energy communities is in a formative phase, we believe it necessary for practitioners and researchers to engage in how collective energy futures may be socially anchored in diverse contexts before emergent technology is embedded in everyday life.

In this article, we draw on five design cases to illustrate the nuances of emergent energy communities. Each design case explores different approaches to design for green community technology that are evaluated in the places people live. By highlighting technology-oriented visions and human-centered expectations of energy communities, we hope to better understand their transformative potentials and pitfalls.

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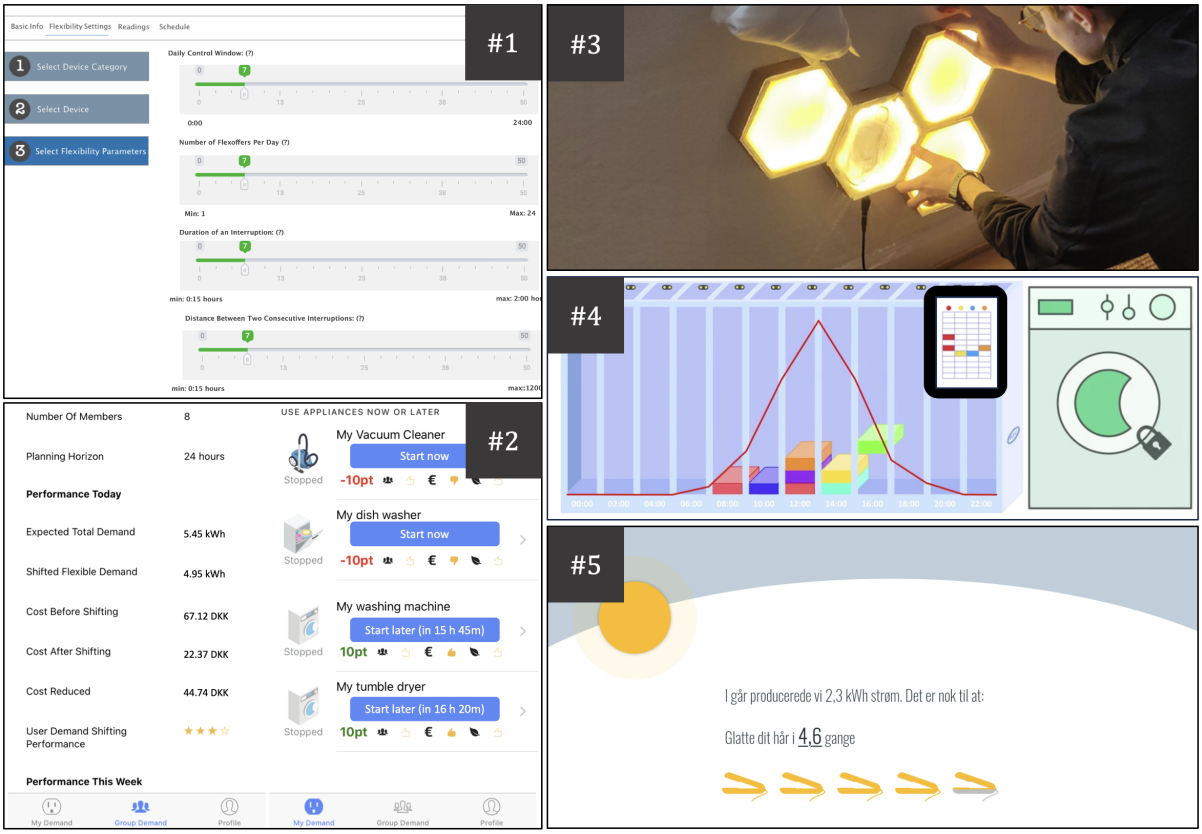


FIGURE 1. Five design concepts. **Top left**: High-fidelity prototype of a user interface for **the GOFLEX HEMS (#1)** trading platform. Here, a householder is asked to select various flexibility parameters, helping to determine the automated operation of their heat pump. **Bottom Left**: **The Community Energy Planner App (#2)** illustrating different time recommendations for running household devices and the (dis)incentives. The app also provides information on the overall collective performances of the energy community. **Top right**: **The LUMEN Ambient Home Display (#3)** mounted on the wall of a participating household. Each hexagon light represents a household in the community giving feedback on how the community currently is using green local energy. **Middle right**: **The Laundry Planning Game (#4)** is a collective planning "game" inspired by the game "Connect Four'. In this game, participants are expected to align laundry times to when a housing cooperative produces solar energy. **Bottom right**: **The Energy Metaphors Public Display (#5)** illustrates the time of the day (the sun over the horizon) a particular activity (here, a flat iron representing personal care) is most likely to take place. Using energy data from yesterday's local solar production, the colored flat irons illustrate the number of times a single student can engage in this activity (here, 4.6 times).

Design Case Background

To understand the challenges and potentials surrounding technology-oriented visions and human-centered expectations of energy communities, we draw on five design cases that we conducted related to the concept of energy community and communal and regional energy. These cases involved different partners and stakeholders across energy research and industries, including computer science (CS) researchers, human-computer interaction (HCI) researchers, software and hardware developers, energy utility companies, as well as participating local communities and households.

In all the design cases, Author 1 played a central role in shaping human-centered design and evaluation efforts, either as an active HCI research partner in various EU-funded research projects [3], [7] or exploring environmental and human-centered design concerns together with HCI and CS colleagues and students (including Author 2) [4], [5], [6]. Author 3 helped to critically engage with articulations of the sustainable sociotechnical futures narrated in the five design cases. In the following, we briefly describe the five design cases (see table 1 for a summary) before

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TABLE 1. Five design cases related to energy communities.

Design Cases	Pervasive Energy Technologies	Settings	Empirical Methods
Design Case 1 **The GOFLEX HEMS** [3].	Local energy flexibility markets, aggregator, flexibility modeling, home energy management system (HEMS), home automation	Households in rural town settings in Germany, Switzerland, and Cyprus	Household survey with 167 participants. A content analysis of developers' assumptions
Design Case 2 **The Community Energy Planner App** [4].	Energy recommendations, energy efficiency models, energy data prognosis, collective energy goals, e.g., saving money or using green energy	Virtual energy communities, Danish households, unknown community members	Field deployment study with six families, pre- and post-interviews
Design Case 3 **The LUMEN Ambient Home Display** [5].	Ambient light home display, smart, digital application, community energy consumption data, smart plugs	Socially acquainted groups, Danish households, virtual energy communities	Field deployment study with three families, pre- and post-interviews
Design Case 4 **The Laundry Planning Game** [6].	Design concepts, community-oriented eco-feedback, practice-based game, smart plugs	Spanish housing cooperatives	Focus group session
Design Case 5 **The Energy Metaphors Display** [7].	Public eco-feedback display, aggregator, energy sector coupling, solar production data, energy metaphors	Small village in Denmark, households, local companies, public organizations	Value-Sensitive Design, contextual inquiry, co-design workshop, interviews, desk research

articulating relevant themes and concerns across the five design cases.

Design Case #1, the GOFLEX HEMS project, concerned the exploration and development of European future green local energy markets. In this technology-driven project, participating households were introduced to different developed home energy management systems (HEMS), trading "energy flexibility" in a local green energy market on participants' behalf (see figure 1:#1). The HEMS and the developmental local energy market were tested and evaluated in three rural areas in Germany, Switzerland, and Cyprus. User evaluations involved a survey study covering householders' motivations, experiences, and future expectations of HEMS technology and local flexibility markets. We further explored the project partners' assumptions of the developed technology and household participation.

Design Case #2, the Community Energy Planner app, involved a field deployment study with six Danish families. The app (see figure 1:#2) was developed by a local smart-grid technology start-up company and allows households to gather in virtual energy communities working towards a collective energy goal. The system offers recommendations on when to do household practices with embedded energy-consuming devices, such as washing and drying clothes, vacuum cleaning, and dishwashing. The recommendations are based on energy efficiency models, energy data prognosis, and individual and collective consumption patterns. We studied the families' interactions with the app and their experiences of following a collective energy goal.

Design Case #3, the LUMEN Ambient Home Display, concerned the design of a community-oriented physical home display deployed with three Danish households. The LUMEN home display was designed as a grouping of physical hexagon lights (see figure 1:#3), representing households in the energy community. The ambient lights were mounted on the wall at a central location within the participating households, and interaction was supported by a digital application allowing householders to explore community consumption data captured by smart plugs monitoring domestic devices. This design aimed to give immediate eco-feedback on collective energy consumption.

Design Case #4, the Laundry Planning Game, involved different community design concepts targeted at Spanish energy housing cooperatives interested in investing in solar panels in the near future. One of the design concepts was purposely designed as a trigger for targeting the practice of doing laundry. We based this design on a collective planning "game" inspired by the game "Connect Four" and how public laundromats are typically organized (see figure 1:#4). The purpose of the laundry planning design was for members of a Spanish energy community to schedule their washing machines according to local green energy production. The designs were evaluated in a focus group session with members of different housing cooperatives.

Design Case #5, the Energy Metaphors Public Display, involved Denmark's first citizen-led ener-

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gy community, EnergiFællesskab Avedøre (EFA). The nexus of the EFA energy community is situated at the local high school, which also houses the local utility company and community-owned solar panels. In collaboration with a software company and the above partners, we were tasked to design a public eco-feedback design to be displayed at the local high school. To bring a human-centered perspective on the design of energy production information stemming from the high school's solar panels, we conducted a co-design workshop with 36 students to design different energy metaphors based on students' reflections on energy in their everyday lives. These energy metaphors were implemented in the public eco-feedback display (see figure 1:#5).

Based on the learnings across the five design cases and CS/HCI literature, our aim here is to articulate a set of design-relevant themes and tensions to form the basis for future research and the design of technologies for energy communities that are more inclusive, meaningful, and sustainable. We situate this articulation within the practice-oriented paradigm, which highlights that (energy-intensive) performances are *"situated in time and space, and are dependent on many features of the surrounding material and cultural environments [...] interwoven within a practice"* [8], and that such everyday practices shape the relations between people, energy consumption, and pervasive energy technology [9].

Firstly, we present three themes related to technology-oriented visions surrounding communal energy technology. Secondly, we point to three human-centered expectations of collective energy futures. Lastly, we articulate future research trajectories that better account for the diverse social lifeworlds of energy communities.

Design Visions for Pervasive Communal Energy Technology

Three overarching themes emerged from the five design cases concerning the design visions embedded in pervasive energy technology for energy communities. The themes, described below, capture key aspects of how designers, developers, and engineers envision the technical, computational, and designerly feasibility of energy community technologies for enabling local green energy futures.

Datafying Communal Energy

In the past decade, we have seen various emergent technologies showcasing the capabilities of digitizing "invisible" energy resources to help datafy energy consumption and green production patterns. This includes our five design cases where we witnessed the roll-out of domestic smart meters, home management energy systems (HEMS), and smart plugs for domestic appliances, each collecting various aspects of domestic energy data. These pervasive technologies are capable of capturing rich energy data patterns [10] i) energy data *granularity* (from a national, regional, neighborhood, household, down to individual device level), ii) energy data *metrics* (CO_2 emission, kWh, kW, dynamic price signals, grid congestion, energy flexibility, and green energy production), and iii) at different *frequencies* (historical, near real-time and forecasted energy patterns). This datafication of energy patterns has enabled researchers and designers to explore ways to represent materialized "invisible" energy in interactive eco-feedback designs [11]. With a vision to create green awareness, these technologies are designed on the assumption that energy data will provide a rationale for the individual householder to change energy behaviors [12].

Our design cases illustrate possibilities to move beyond individualized datafication toward more collective understandings of energy data. For instance, the Energy Community Energy Planner App (#2) exemplifies how predictions of future energy patterns are used to compose collective energy recommendations. In this design case, recommendations are assembled as chosen community "goals", which can be articulated by community members as "saving money" or "saving the environment". How forms of tangibility and aesthetics mediate energy data through design is illustrated with the LUMEN Ambient Home Display (#3). Here, the physical wall-mounted light display was purposely designed to provide a pervasive, ambient presence to remind householders of their relations to the energy community. To better situate energy data within the local energy community, the Energy Metaphors Display (#5) highlights how energy data may be used to co-design energy metaphors with students of a community. These energy metaphors attempt to translate incomprehensible energy metrics (kWh, CO_2 emissions, energy prices, etc.) to relatable energy-consuming practices performed in these students' everyday lives (body care, going to school, cooking with the family, and commuting).

These cases exemplify numerous design choices that exist for designing with communal energy data that render intangible and incomprehensible energy consumption and production patterns. Yet, at the same time, most energy data capture objective energy metrics, meaning these designs reframe assump-

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tions about how data might engage the individual and yet community-oriented energy consumer akin to Strengers' Resource Man [12], a tool empowered user that is motivated by data-driven rationalized logics. Nevertheless, our studies show that such data rarely capture the social rhythms and cultural practices situated in local communities. We will discuss these concerns in later themes.

Decentralizing Energy Market Infrastructures

As we become more reliant on locally situated renewable energy technologies (wind turbines and solar panels), the availability of energy fluctuates along with unpredictable weather conditions. To overcome such uncertainty in the energy grid, digital energy technology is transforming our underlying energy infrastructure to better align local energy demand to when and where green energy is produced. This transformation has led to the emergence of novel energy markets, where local actors are expected to engage in community energy trading [13]. The European Union (EU), for instance, has a stated vision of constructing digital infrastructures inviting interested actors to participate in decentralized energy markets, making it more straightforward for citizens to form energy communities [1]. In these future energy markets, pervasive energy infrastructures (e.g., micro-grids, smart-grids, blockchain, and Peer-to-Peer energy markets) are envisioned to automate the coordination, production, consumption, and distribution of local green energy [14].

Such decentralized energy infrastructures, also known as "bottom-up energy systems" [3], are envisioned to empower householders to participate actively in trading different types of energy in the local community, leading to more just, democratic, and sustainable energy futures. The GOFLEX HEMS (#1) explores the potential and challenges of trading local "energy flexibility". This market infrastructure implies that householders are willing to trade "energy flexibility" of specific domestic devices (e.g., washing machines, heating and cooling devices, fridges, computers, and EVs). For instance, the energy flexibility of a heating device in a household can be computationally modeled, commodified, and traded [3] based on the device's "optimal" running time. In this case, the commodified value of energy flexibility is determined by the householders' willingness (or not) to align consumption with production, e.g., heating their home when local, green energy is produced. The vision is that this value somehow represents compensation for the "inconveniences" and "discomforts" the household may have encountered. Similar market mechanisms are captured in the collective energy recommendations in the Community Energy Planner app (#2). Here, stating willingness to align device consumption to the "right" time is being envisioned to be financially compensated as well as awarded with positive "points" counting toward the collective "goal".

While it is technically possible to capture and model energy flexibility as a commodity to be traded in smart energy communities, our design cases illustrate that engaging householders in such markets may not be as straightforward as envisioned. Socio-economic gains may be rationalized as an effective way of envisioning householders engaging with "bottom-up energy systems". Yet, our findings show that other social desires and expectations (e.g., "being it together", feeling included and culturally represented) shape meanings of communal energy. Hence, such rationalized market values those "green" markets are based upon [15] often fall short of capturing the social and cultural desires that may form future sustainable communities.

Algorithmic Energy & Household Automation

Since Weiser and Brown coined the vision of calm, pervasive computing, computational models and algorithmic automation have become embedded in most aspects of everyday life. This includes visions of energy technologies where ideas of calm computing currently shape the development of domestic energy systems targeting local green energy production.

In the GOFLEX HEMS (#1) project, the home energy management system embedded visions of algorithmic automation that would "ensure" minimal household interaction and individual household expectations were not breached by the system (e.g., keeping a comfortable temperature in the home, having a charged car when needed). To help capture such "human complexities", built-in algorithms continuously learned a household's energy "strategies and behaviors" (learning when a household may like to wash clothes or charge their EV) and align such expectations with local energy production. To give the householders a sense of human agency, interactive "parameters" were designed to allow householders to "take control" when needed (see figure 1:#1).

In the same case, *aggregators* were introduced as valuable partners to the local energy system, as they may provide computational models and algorithmic automation to help smaller energy producers to sell, buy, and distribute energy and energy flexibility on decentralized energy infrastructures [3]. Findings from our study do show that householders were intrigued by this "innovative" system operating in their homes. However,

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most participants also indicated they were confounded by how the system "managed" their "green" homes. For instance, some experienced unnecessary cooling or heating times. While the system interpreted such actions as "sustainable" as running times of the devices were scheduled for green production hours, participants understood these actions (and understandably so) as unsustainable consumption patterns.

Computational models also shaped the energy recommendations in both the Community Energy Plan ner app (#2) and the LUMEN Ambient Home Display (#3), assuming to permit householders to "monitor" and "manage" both individual and collective energy patterns. However, participants often felt that they had to adapt their busy and messy everyday lives to these recommendations and, therefore, found it difficult to incorporate these recommendations into their everyday practices. The Laundry Planning Game (#4) turns the vision of calm computing upside down. Here, the design purposely provokes and makes energy-intensive laundry practices more difficult to perform. Focusing on triggering reflections through design in a playful and meaningful way, we based this design on a collective planning "game" inspired by the game "Connect Four" and how public laundromats are organized. In this design concept, smart plugs only turn on washing machines if the Spanish community collective manages to schedule laundry time after communal solar production.

Our cases show that algorithmic automation of energy technology is typically envisioned to serve non-intrusive agency while calmly blending into everyday life. These studies seem promising, but we also suggest focusing on reimaging how algorithmic automation design may shape more meaningful and engaging collective experiences, as also suggested by [16].

Participating in Collective Energy Futures

In the vision of energy communities lies expectations that local, small-scale, and collective approaches will sustain engagement and participation. Yet, how supporting energy technologies are to be socially and politically anchored in the everyday life of these diverse communities is rarely considered. In the following themes, we reflect on the householders' experiences as reported in our five case studies.

Engaging the Householder

In the five design cases, different design techniques were employed to engage householders in the shared understandings of how an energy community may collectively act upon sustainable transitions. Across our studies, findings suggest that feeling part of a community that collectively cares and cooperates with sustainable challenges fosters a more meaningful understanding of local green energy initiatives than attempting to consume green energy on an individual basis. Indeed, the Community Energy Planner app (#2), the Laundry Planning Game (#4), and the Energy Metaphors Public Display (#5) cases illustrate that collective action points of energy communities become more telling in already established social communities. It also became apparent in the LUMEN Ambient Home Display (#3) that energy technology alone cannot establish relationships between unknown members that aim to foster collective experiences.

Illustrated in our design cases, a key design aspect of fostering collective experiences is to materialize community energy data in interactive designs. In the design cases (#2) and (#3), energy data were used to shape collective experiences framed as collective "green" goals. These goals were made achievable in the individual household by following green consumption recommendations (using local green energy at the "right" time). Making individual compromises of convenience and comfort was at first seen as something that led to "the greater good for the community". However, partaking in goal-oriented energy recommendations in a busy everyday life led to nuanced experiences. Many householders turned consuming energy into a game. While some participants found it fun to "win", others reflected that competition elements were disengaging. Others described feelings of guilt and shame associated with being part of a community that collectively is measured on individual performances, particularly when individual actions are designed to influence the community's collective "goal" in a "negative" way. Of particular importance, we found competition mechanisms led to some participants becoming more focused on being "the best" household, moving away from the collective understanding of "being in it together" or acting "green". We witnessed that such social dynamics in the community have unintended consequences that can lead to unsustainable practices. For instance, some households chose to use the dryer in the middle of the night to "score points" by using green energy rather than drying clothes outside, which was normally done in many of the households.

These findings show that visions of socio-economical gains and goal-led recommendations hardly capture the messiness of everyday life nor question how design may challenge the status quo. Instead, the design rationale of framing green commu-

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nity awareness and togetherness through such game mechanisms ended up, for many of the householders, driving unsustainable competitiveness and fostering feelings of guilt and unpleasantness. Indeed, such individualized game mechanisms may end up encouraging householders to perform energy-intensive actions (a household drying clothes in the dryer instead of doing it outside as "normally" done to score "points"). This implies a shift to the Social Man, a close relative of the Resource Man, who is motivated by sharing and comparing his energy actions with other Social men [12]. However, this shift is insufficient to support green communal energy transitions [4]. To move beyond such individualized framings, our work in the Laundry Planning Game (#4) and the Energy Metaphors Public Display (#5) suggests we need better ways to embrace the situated social aspects of everyday life that also foster long-lasting communal participation [17], [18].

Participating in Energy Communities

Collective, democratic, and local participation is one of the core arguments of the EU to bring political support for establishing energy communities [1]. Yet, embedded in visions of European energy communities is an expectation that sustainable, democratic futures will naturally follow without fully embracing what just, democratic, and sustainable futures might look like situated in community sites.

Our design cases illustrate the complexities of understanding and designing for participation in energy communities. The GOFLEX HEMS (#1), the Community Energy Planner app (#2), and the LUMEN Ambient Home Display (#3) demonstrate that technology-oriented visions assume and predefine democratic participation in energy communities as "things" that will naturally follow with "neutral" technology being embedded in these communities. Participation in these cases is often envisioned to be financially motivated, where algorithmic technology will ensure fair trading and just access to local energy, while automated home technology will "manage" the responsibility of participation [2], [13], [14]. Yet, our findings imply that when these technologies become too pervasive and distant from the situated places where participation is imagined and unfolded, it is unclear for participating householders how transformative green energy efforts are to be pictured and achieved [3]. Hence, we see that when other actors and technologies (aggregators, HEMS, energy traders) are provided with strong agency, questions about participation move away from a human-centered focus, leading to disengagement and human-omitting participation.

At the same time, our design cases illustrate that most of these infrastructures, algorithms, computational models, energy markets, and "bottom-up energy systems" are developed by engineers in a traditional top-down development fashion. Overall, our findings illustrate that participation in energy communities brings about many nuanced complexities hardly ever captured in "one-size-fits-all" solutions. Instead, these computational models and algorithmic automation are typically envisioned, designed, and developed by, e.g., engineers and stakeholders from the energy sector, far removed from diverse communal contexts where people are living here. The various design cases demonstrate that participation in these local communities (in the EU) is performed differently and often has different meanings. For instance, rules to govern democratic participation in Spanish housing cooperatives [6] or the century-old Danish cooperative movement shaping Denmark's first citizen-driven energy community [7] already exist and have been formed over time and places. Furthermore, visions of the empowered householder and democratic community participation are typically shaped by stakeholders from the industry [3], [16] and policymakers [1]. Yet, as our design cases show, notions of energy justice and democratic engagement are shaped socially and culturally by the people living in these communities [7]. As these communities are diverse, so are their understandings and imaginings of a just, sustainable future.

These findings highlight that the predominant top-down approach to developing energy infrastructures and computational models for energy communities may overlook the entangled complexities and meanings that arise from direct "bottom-up" design involvement in these local communities. Across the EU (and beyond), local communities display diverse everyday life dynamics, which impact their understanding of green energy consumption. While industry stakeholders influence ideas of the empowered householder and democratic involvement, our design cases suggest that we need better ways to understand and frame participation, by designing together with these diverse communities [18].

Situated Community Lifeworlds

Studying each of the design cases in situated contexts highlights how social and cultural dimensions shape communal life (including energy consumption) in diverse ways. From the GOFLEX HEMS (#1), the Laundry Planning Game (#4), and the Energy Metaphors Public Display (#5) cases, we came to recognize how energy community participation is embedded in, and

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even centered around, local culture. Other studies [17], [16], including our Community Energy Planner app (#2), and the LUMEN Ambient Home Display (#3) cases, have sought to bring together diverse households in energy communities. In our design cases, we see i) *design techniques* (eco-feedback, collective goal recommendations), ii) *decentralized energy infrastructures* (bottom-up systems, local trading markets, aggregators), and iii) *algorithmic automation* (HEMS and energy monitors) all attempting to create coherent community experiences for participants. However, these vision contains a "one-size-fits-all" expectation that geographically, culturally, and domestically different households have the same desires and expectations for communal energy living.

This assumption was challenged in most of our design cases. For instance, keeping a comfortable temperature at home had different meanings in the local communities participating in design case (#1). In Cyprus, this often meant cooling your home, while in mountain villages in Switzerland, comfort expectations called for heated homes in the winter [3]. In design case (#5), we also found that students were expected to use aggregated local data from EFA to learn about sustainability and energy, as well as foster local democratic participation in Denmark's green transition [7]. This is a locally anchored value, where the presence of EFA's nexus at the high school led to this consideration. During our workshop in EFA (#5), we saw the diversity of practices illustrated when we asked students to illustrate different everyday energy-consuming and producing actions. Here, we found two things striking. First, actions included plane travel. Many of our participants were adult immigrants living in Denmark and getting a high school diploma to continue on to Danish university. The plane travel action illustrated how they traveled to see family in other countries. Similarly, many workshop participants were parents leading to a number of illustrations of washing machines for washing children's clothes.

We believe these findings make a case for advocating for bringing *pluriverse* perspectives into designing with energy communities, where diverse communities in energy engagements can coexist [19], rather than a homogenization of energy communities across local, social, and cultural lifeworlds. Such pluriversal bottom-up approaches call for practitioners to reshape their generalizable community visions to better encompass energy-consuming practices in the local communities, their different social rhythms, and even their non-human entities, e.g., the surrounding nature, non-human beings (different animal and plant species), and other physical and digital "things".

Next Steps

Above, we have articulated a number of design-related themes and tensions regarding the energy community participation and design vision of pervasive energy technology. The key takeaways from our learnings across the five design cases and CS/HCI literature are summarized in table 2. We hope these may form the basis for future research and design of technologies for energy communities. We now discuss how such future research trajectories can be unfolded.

Transdisciplinary Engagements

Overall, our case studies illustrate the imagination of the energy community is mostly shaped by developers, engineers, and energy stakeholders, often, far removed from the political, social, and local lifeworlds where the energy technology solutions are intended to co-exist. In these scenarios, "engineers" frequently envision future energy community participation using computational modeling and measurable energy data. While such quantifiable approaches may help qualify how alternative green community life might look like, they may also create a gap between research concerns, political development, and aspirations for transformative sustainable changes [15].

To nurture alternative visions of energy communities, we see great value for interaction designers, social scientists, and software developers to engage in transdisciplinary collaborations centered around local communities. For example, social scientists and interaction designers can help identify possible tensions early in the energy community design process, while engineers can identify possibilities for green energy data collection and democratic algorithmic concerns. In our work, we have seen wishes for such transdisciplinary engagements to help qualify developed solutions. Our experiences tell stories that such arrangements bring nuanced perspectives to the development and design rationales embedded in the design of pervasive energy technology – and those nuanced perspectives could be a way forward for challenging technology-oriented visions and assumptions of human participation within these.

Therefore, we consider it relevant for researchers and practitioners alike to engage with more critical, speculative perspectives of communal energy to move beyond visions of calm computing. Such design approaches may better embrace social aspects of everyday life by triggering and provoking energy-consuming practices through design [6]. In such approaches, HCI researchers are specifically equipped to engage with technology-oriented engineers to collaboratively

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TABLE 2. Key takeaways across the five design cases and HCI/CS literature.

Design Visions for Pervasive Communal Energy Technology	• **Datafying** "invisible" energy patterns via objective energy metrics and materialized through gamification mechanics, reshapes assumptions about the rational and community-oriented energy consumer. ([10], [11], [5], [12]). • **Decentralizing** energy infrastructures as, e.g., flexibility energy trading, assumes socio-economic gains may be rationalized as an effective way of envisioning engagement with "bottom-up energy systems". Yet, such rationalized logic may fall short of embracing culturally diverse lifeworlds. ([15], [3], [4], [13], [14], [17]). • **Algorithmic automation** of energy technology is typically envisioned to serve non-intrusive agency, calmy blending into the background of messy everyday life. ([3], [9]).
Participating in Collective Energy Futures	• **Engaging** householders through rationales of game mechanisms to bring about green community awareness and togetherness may drive nonsustaining competitiveness and feelings of guilt and might actually lead to more energy-intensive practices. ([4], [9], [15]). • **Participation** in energy communities is normally framed by top-down approaches and computational models, which may overlook the entangled complexities and meanings that can arise from "bottom-up" design involvement in local communities. ([3], [17], [7]). • **Situating** community lifeworlds in local contexts call for pluriversal bottom-up approaches that better encompass social and cultural practices in the local communities, their different social rhythms, and their non-human entities. ([18], [6], [19]).

reshape the meanings and expectations we ascribe to energy community systems [3], [9]. Hence, we believe transdisciplinary engagements may not only trouble socio-economic perspectives of being human but also meanings of democratic, just energy community life.

Reimaging Communal Energy Futures

We suggest that as a community engaging with researching and developing future pervasive energy technologies, we need to explore alternative ways of developing, designing, and imagining participation in energy communities to better challenge the status quo. For instance, the development of "bottom-up energy systems" also calls for bottom-up design approaches. For these systems to matter, it is critical that communities, rather than being invited into top-down, established energy community projects, are integrated from the bottom-up. Further, we suggest moving beyond designing for the individual, rational community member [12] and instead focusing on how "bottom-up" design approaches can help sustain engagement and challenge local practices in ways that also emphasize change at political levels [18].

To understand and capture the bottom-up messiness of everyday life, we can draw from ongoing research fields of participatory design, design anthropology [18], and pluriversal design [19]. We believe such approaches are useful for unfolding new visions of participation in energy communities by empowering these communities to create alternative sustainable futures. One approach could be to frame future energy communities as participating *commons* [20], where access and caring for shared local resources, like local green energy, are shaped by collective meanings and action. Such commons could even be supported by digital technology like blockchain [16]. At the same time, we believe that by understanding energy communities as entangled social and technical configurations [8], [17], we are better equipped to embrace the plurality of how these different practices play out in everyday social life. They may even help us move towards more-than-human energy communities where other beings and "things" are also represented in design. Such approaches may then help critique and question to what degree technologies should support current ways of "being human" [19].

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